



encoding

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1 Introduction

Understanding critical combustion events is a complex task that involves extensive analysis of both experimental and numerical data. Finding patterns in these datasets is an important step towards building better models for combustion phenomena. Many tools exist both in the reactive and non reactive flows. Proper Orthogonal Decomposition (POD) (CITAZIONE), is an Singular Value Decomposition (SVD) based method that provide a low rank representation of the flow rewriting it as a sum of orthogonal spatial patterns. Principal component analysis has been successfully employed to project high-dimensional data onto a lower-dimensional orthogonal subspace of thermodynamical variables (CITAZIONE), effectively reducing the complexity of the system while retaining its essential features. However, this method is not able to capture the temporal evolution of the system and it is not able to provide a physical interpretation of the modes. Dynamic Mode Decomposition (DMD) (CITAZIONE) allows the reduction of the system in spatio-temporal modes which have both a spatial structure and an associated temporal frequency and growth rate. Thus the system is represented as the sum of modes evolving with a linear dynamics in time that allows the study of the spectral properties of the system. Although powerful this method suffers from noisy data and it often is inadequate to provide an accurate description of turbulent reactive flows. High Order Dynamic Mode Decomposition (HODMD) (CITAZIONE) is a extension of DMD that improves the method by introducing the higher order Koopman hypothesis. This makes it more robust and has already been successfully applied in reactive flows (CITAZIONE) Combustion data often presents itself in tensorial form. Moreover the most valuable datasets, i.e. the ones that most closely represent the real physics of the system, are often the ones with the highest dimensionality both in spatial component and in the number of variables. This has sparked interest in methods that process data directly into this form and don't require reshaping into matrix form as in the case of SVD based methods and DMD.

The current report focuses on describing the application of these techniques to enhance the algorithms that allows pattern identification in combustion data. High Order Singular Value Decomposition (CITAZIONE), a generalization of SVD, is discussed as a tool to both compress and extract physical patterns from huge combustion datasets. In particular, this method allows for multi modal analysis along each axis of the tensor at the same time, allowing to quantify the importance of each dimension of the tensor. When combined with HODMD this, allows for identification of key feature in the dynamics of the reactive flow.

Furthermore,

2 Datasets

The methodologies are assessed on different datasets from the consortium. These data describe high fidelity simulations of complex reacting flows.



2.1 NH₃/H₂ non-premixed mixing layer

This dataset consists of direct numerical simulations (DNS) of a two-dimensional non-premixed ammonia-hydrogen-air reacting mixing layer. It is generated by DCGIOVANNI at CORIA using the structured mesh reactive flow solver SiTCom-B [6]. The configuration represents a splitter-plate stabilized flame in which a cold fuel stream and a hot air stream interact across a shear layer, representative of conditions encountered in practical burners.

The lower stream consists of a NH₃/H₂ fuel blend (with 15% H₂ by volume) at ambient temperature (300 K) with a mean velocity of 15 m s⁻¹, while the upper stream consists of preheated air with a mean velocity of 39.6 m s⁻¹. A cold wall maintained at 600 K is imposed at the top boundary, and a symmetry condition is applied at the bottom. Homogeneous velocity fluctuations of 10% are prescribed at both inlets. The computational domain measures 12 × 1 cm and is discretized on a uniform grid of 2400 × 200 cells with a resolution of 50 μm.

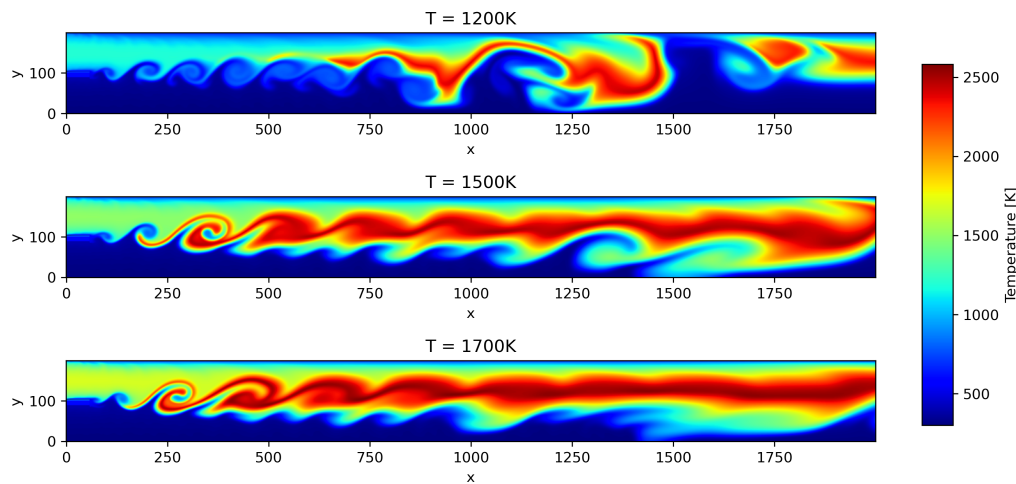


Figure 1: Three representative snapshots of the temperature field from the NH₃/H₂ non-premixed reacting mixing layer DNS, illustrating the spatial evolution of the flame and mixing structures at different instants.

Three datasets are generated at different air preheat temperatures, namely 1200 K, 1500 K, and 1700 K, to probe the sensitivity of the flame dynamics and species evolution to the inlet thermal boundary condition. The domain geometry, grid resolution, and flow velocities are identical across all three cases. Each snapshot contains 19 reactive scalar fields corresponding to the species of the reduced mechanisms RmAH-1 and RmAH-2 [3] — namely H₂, O₂, H₂O, H, O, OH, HO₂, NH₃, NH₂, NH, N, NO, NO₂, N₂O, HNO, NNH, N₂H₂, N₂, and Ar — together with temperature, yielding 20 variables in total. The species analysis presented in this work is conducted exclusively on the 1500 K dataset, which constitutes the reference operating condition for the assessment of the proposed methodologies.

3 Methods

3.1 High Order Dynamic Mode Decomposition

Dynamic Mode Decomposition (DMD) [7] is a data-driven spectral method that extracts spatially coherent structures and their associated temporal dynamics directly from time-resolved snapshot¹ data. Given a sequence of snapshots $\mathbf{v}_k \in \mathbb{R}^J$, $k = 1, \dots, K$, DMD seeks a linear operator $\mathbf{A}$ such that $\mathbf{v}_{k+1} \approx \mathbf{A} \mathbf{v}_k$, whose leading eigenpairs encode the dominant frequencies and growth/decay rates of the system. The full data field is then expressed as a superposition of DMD spatial modes $u_p(\mathbf{x})$ with purely exponential temporal evolution,

$$v(\mathbf{x}, t) = \sum_{p=1}^P a_p u_p(\mathbf{x}) e^{(\delta_p + i\omega_p)(t-1)\Delta t}, \quad (1)$$

where $a_p \geq 0$ are real amplitudes, ω_p are angular frequencies, δ_p are growth rates, and Δt is the sampling interval. Each mode is normalised so that its root-mean-square norm equals unity.

Higher-Order Dynamic Mode Decomposition (HODMD) [4] extends DMD making it more suitable for noisy data and nonlinear phenomena such as the ones encountered dealing with combustion data. The main idea is to extend the assumption that the system's evolution can be described with a linear operator to temporal windows of size d . Each snapshot is thus related to the d subsequent snapshots through a set of linear operators. This generalised assumption leads to a sliding-window construction of an augmented snapshot matrix and allows for better identification of close frequency even in noisy data. When $d = 1$, HODMD reduces exactly to standard DMD.

The HODMD algorithm proceeds in two main steps:

Step 1 — Dimensionality reduction via SVD. A truncated Singular Value Decomposition (SVD) is applied to the original snapshot matrix, compressing the spatial dimension from J to N retained singular values. The tolerance ε_{SVD} controls the level of compression by enforcing

$$\frac{\sigma_j}{\sigma_1} \geq \varepsilon_{\text{SVD}}, \quad (2)$$

where σ_j are the singular values in decreasing order. The resulting temporal coefficients weighted by the retained singular values form the reduced snapshot matrix, which filters noise.

Step 2 — High-order Koopman eigenvalue problem. HODMD is applied to the reduced snapshot matrix with delay parameter d . The eigenvalue problem is solved to extract DMD modes, frequencies and amplitudes. A second tolerance ε_{DMD} selects physically significant modes by retaining only those whose relative amplitude satisfies

$$\frac{a_p}{\max_p(a_p)} > \varepsilon_{\text{DMD}}. \quad (3)$$

¹Matrix that has as columns (rows) the time index and all the other unfolded indices as rows (columns)



3.1 Physical interpretation

From a practical point of view DMD looks for coherent structures that behave in an oscillatory manner across the whole dataset. This makes it easier to observe phenomena that cannot be observed when looking at the original data, that often is very complex. A big issue is that the assumption that the system can be described with a matrix even in snapshot form is very strong and it will fail in case of noisy or turbulent data. HODMD is able to obtain better results because instead of looking for oscillatory coherent structures across the whole dataset, it looks for structures that depend on the last d snapshots instead of just the last one. This makes it possible to extract up to $d \times N$ modes instead of N :

$$\underbrace{\mathbf{v}_{k+1} = \mathbf{A} \mathbf{v}_k}_{\text{DMD}} \longrightarrow \underbrace{\mathbf{v}_{k+d} = \mathbf{A}_1 \mathbf{v}_{k+d-1} + \mathbf{A}_2 \mathbf{v}_{k+d-2} + \dots + \mathbf{A}_d \mathbf{v}_k}_{\text{HODMD}}$$

3.2 Multidimensional Iterative HODMD (mdHODMDit)

3.2 High Order Singular Value Decomposition

Both DMD and HODMD perform truncated SVD for denoising and dimensionality reduction. This effectively filters out structures that have very low temporal contribution across the whole dataset, which often coincides with noise. Combustion data often carry several physical variables and span a multi-dimensional spatial domain. The application of SVD on the reshaped tensor introduces an implicit coupling among modes that belong to different dimensions, and the SVD truncation cannot distinguish noise contributions arising from the spatial, temporal, or variable axes independently. For this reason, a first improvement over the classical version of HODMD is to introduce High Order Singular Value Decomposition (HOSVD) [2] in the first step. This is a generalization of SVD that allows filtering along each dimension of the tensor.

It works on data keeping it in tensor form $\mathcal{T} \in \mathbb{R}^{N_{\text{var}} \times N_x \times N_y \times N_t}$. First, it unfolds $\mathcal{T}$ along each of its four modes in turn, computes the n -mode singular values $\{\sigma_j^{(n)}\}$ from the corresponding unfolded matrix, and retains factor matrices $\mathbf{U}^{(n)}$ by enforcing the truncation criterion

$$\frac{\sigma_j^{(n)}}{\sigma_1^{(n)}} \geq \varepsilon_{\text{tol}}, \quad (4)$$

independently for each mode n . The result is a compressed core tensor $\mathcal{S}$ and a set of factor matrices $\{\mathbf{U}^{(n)}\}_{n=1}^4$ such that $\mathcal{T} \approx \mathcal{S} \times_1 \mathbf{U}^{(1)} \times_2 \mathbf{U}^{(2)} \times_3 \mathbf{U}^{(3)} \times_4 \mathbf{U}^{(4)}$, where $\times_n$ denotes the n -mode product. These matrices are exactly the ones one would obtain performing truncated SVD on each of the unfoldings. Therefore, the temporal factor matrix $\mathbf{U}^{(4)}$ is the same as before.

After the DMD eigendecomposition, the reduced spatial DMD modes are projected back to the original tensor dimensions by contracting with the spatial factor matrices



$\mathbf{U}^{(1)}, \mathbf{U}^{(2)}, \mathbf{U}^{(3)}$ and the core tensor, recovering physically interpretable spatial structures in the full variable- x - y space without ever vectorising the multi-variable field, from this the algorithm is called mutlidimensional.

Then, the procedure is iterated. The DMD reconstruction is used to form an updated snapshot tensor, which is re-compressed via HOSVD. The spatial ranks r_x , r_y , and r_{var} retained after truncation are monitored: if they differ from those of the previous iteration, the HODMD step is repeated on the new compressed temporal matrix $\hat{V}_1^K$. Convergence is declared when the spatial ranks no longer change between two successive iterations. This can be interpreted as the low-rank spatial structure implied by the DMD modes is the same as the one identified by HOSVD. The result is a decomposition in which the noise filtering and the dynamic mode identification are consistent, rather than performed independently.

4 Results and Discussion

4.1 mdHODMDit Applied to the CORIA Dataset

This section presents the application of the multidimensional iterative Higher-Order Dynamic Mode Decomposition (mdHODMDit) algorithm to the three NH_3/H_2 DNS datasets from CORIA described in Section 2.1. The work is carried out in collaboration with CORIA and builds upon previous developments related to the mdHODMDit methodology, particularly the work of Pablo López Salazar. Within the present study, the computational workflow was applied to the CORIA datasets, leading to the analysis of the results discussed in this section.

4.1 Data Structure and Preprocessing

The DNS output for each temperature case is stored as a four-dimensional tensor $\mathcal{T} \in \mathbb{R}^{N_{\text{var}} \times N_x \times N_y \times N_t}$, where the axes correspond to the reactive scalar variables, the streamwise spatial coordinate x , the cross-stream coordinate y , and time t , respectively.

Before analysis, the tensor is restricted to the region of physical interest. In the streamwise direction, indices $x_{60:1000}$ are retained, corresponding to the range $[0.3, 5.0]$ cm downstream of the splitter plate, which encompasses the near-field flame stabilisation region and the initial development of the shear layer. In time, the first 74 snapshots are discarded to remove the initial transient. The temporal resolution is $\Delta t = 40 \mu\text{s}$.

For this kind of data, scaling is important due to the differences in the order of magnitude of the variables considered and the differences between their units of measurement [1, 5]. Since there is no universally accepted quantitative criterion to determine which scaling method is most appropriate, the analysis was performed using both auto scaling $x_i^{\text{scaled}} = \frac{x_i - \mu_x}{\sigma_x}$ and range scaling $x_i^{\text{scaled}} = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}$. The analysis yielded similar results for both scaling methods, identifying the same main frequencies.



4.1 Parameter Calibration

The analysis is performed over a grid of Hankel delay parameters

$$d \in \{10, 15, 20, 25, 30, 35, 40, 45\} \quad (5)$$

and tolerance values

$$\varepsilon_{\text{tol}} \in \{5 \times 10^{-3}, 2 \times 10^{-2}, 3 \times 10^{-2}, 4 \times 10^{-2}, 5 \times 10^{-2}, 7 \times 10^{-2}, 1 \times 10^{-1}, 2 \times 10^{-1}\}. \quad (6)$$

The same tolerance ε_{tol} governs both the HOSVD truncation (equation (4)) and the amplitude threshold applied to retain DMD modes, so that a single parameter controls the compression level throughout the algorithm.

4.1 Identification of Physical Frequencies

A DMD mode is identified as physical when its spectral peak appears across different d values and it's resilient to the tolerance threshold ε_{tol} . This is due to the fact that the coherent structures exist independently on how the observation matrix is built, thus should be found independently on how the d parameter varies, while keeping in mind the previous discussion. The relative resilience of the peaks with respect to ε_{tol} is because relevant structures should be associated to relevant eigenvectors and not ones close to noise threshold. Frequencies are expressed in units of Strouhal number:

$$St = f \cdot \frac{\ell}{U_b}, \quad (7)$$

where $\ell = 3$ mm is the reference length of the mixing layer and U_b is the bulk speed, respectively $U_b = 21,34, 24,29$ e $27,14$ m s⁻¹ for cases at 1200 K, 1500 K e 1700 K.

4.1 Frequency–Amplitude Spectra and associated DMD modes

The frequency–amplitude plots shown in Figures 2, 4, and 6 display, for each temperature case, the normalised DMD mode amplitudes $|\hat{a}_k| / \max_k |\hat{a}_k|$ as a function of Strouhal number St . Each marker corresponds to one $(d, \varepsilon_{\text{tol}})$ pair; different colours encode different values of d and different marker styles encode different tolerances. Peaks that persist across all curves can therefore be read off directly as physically robust frequencies. To showcase the coherent structures associated to these frequencies, temperature is chosen as the representative variable due to the fact that in many cases is a variable of interest for both experimental and theoretical problems. Despite this, coherent structures for every variable can be plotted.

Case 1200 K The spectrum for the 1200 K case, shown in Figure 2, is characterised by a rich distribution of modes spanning a broad Strouhal range up to $St \approx 1.7$. The dominant peak is located at $St \approx 0.42$ and is reproduced consistently across all $(d, \varepsilon_{\text{tol}})$ combinations, confirming its physical robustness. Secondary peaks of significant amplitude are identified at $St \approx 0.85$, $St \approx 1.1$, $St \approx 1.3$, $St \approx 1.7$. These peaks are approximately integer multiples of the dominant frequency and therefore interpretable as higher harmonics of the fundamental shear-layer instability.



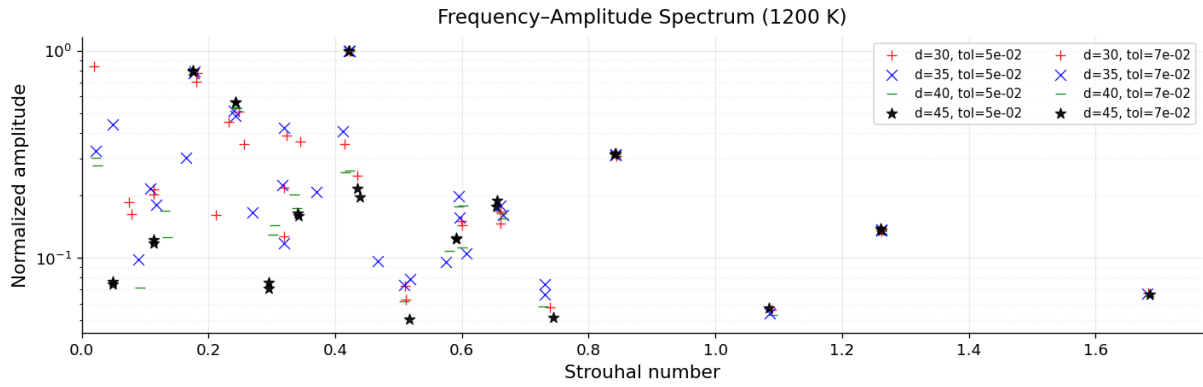


Figure 2: Normalised DMD mode amplitudes as a function of Strouhal number for the 1200 K dataset, for $d \in \{35, 40, 45\}$ and $\varepsilon_{\text{tol}} \in \{5 \times 10^{-2}, 7 \times 10^{-2}\}$.

The associated modes to the highlighted points in Figure 2 are shown in Figure 3.



(a) [PLACEHOLDER] Mode 1



(b) [PLACEHOLDER] Mode 2



(c) [PLACEHOLDER] Mode 3



(d) [PLACEHOLDER] Mode 4

Figure 3: [PLACEHOLDER] DMD modes associated with the dominant peaks in the 1200 K spectrum.

Case 1500 K The 1500 K spectrum, shown in Figure 4, exhibits a structure closely analogous to the 1200 K case, with the dominant peak again located at $St \approx 0.42$. Secondary peaks are found at $St \approx 0.78$, $St \approx 0.95$, and $St \approx 0.1$, $St \approx 1.5$. The overall behaviour is similar to the previous case reported in Fig 2 with the harmonic sequence translated.

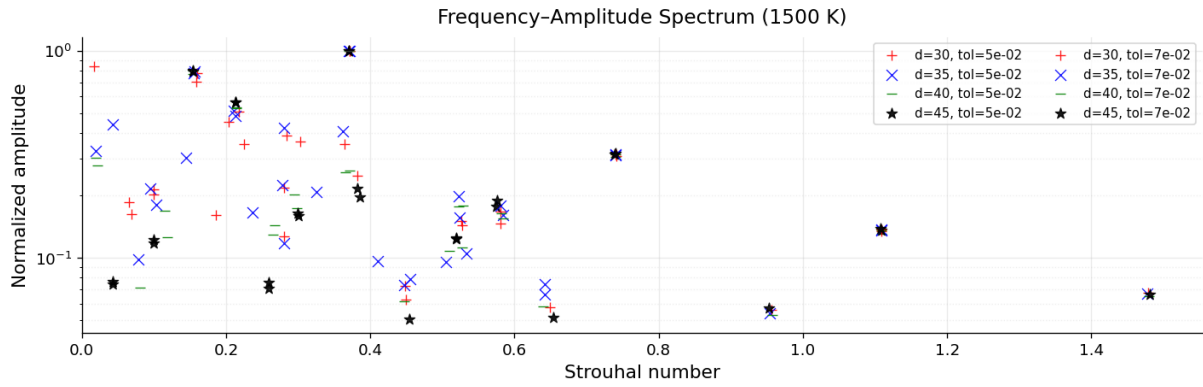


Figure 4: Normalised DMD mode amplitudes as a function of Strouhal number for the 1500 K dataset, for $d \in \{35, 40, 45\}$ and $\varepsilon_{\text{tol}} \in \{5 \times 10^{-2}, 7 \times 10^{-2}\}$.

The associated modes to the highlighted points in Figure 4 are shown in Figure 5.



(a) [PLACEHOLDER] Mode 1



(b) [PLACEHOLDER] Mode 2



(c) [PLACEHOLDER] Mode 3



(d) [PLACEHOLDER] Mode 4

Figure 5: [PLACEHOLDER] DMD modes associated with the dominant peaks in the 1500 K spectrum.

Case 1700 K The 1700 K case, shown in Figure 6, presents a different spectral structure. The dominant peak remains at $St \approx 0.40$, consistent with the lower-temperature cases, but the overall spectrum is sparser: maximum relevant Strouhal number observed is $St \approx 1.25$ and fewer secondary modes are retained.

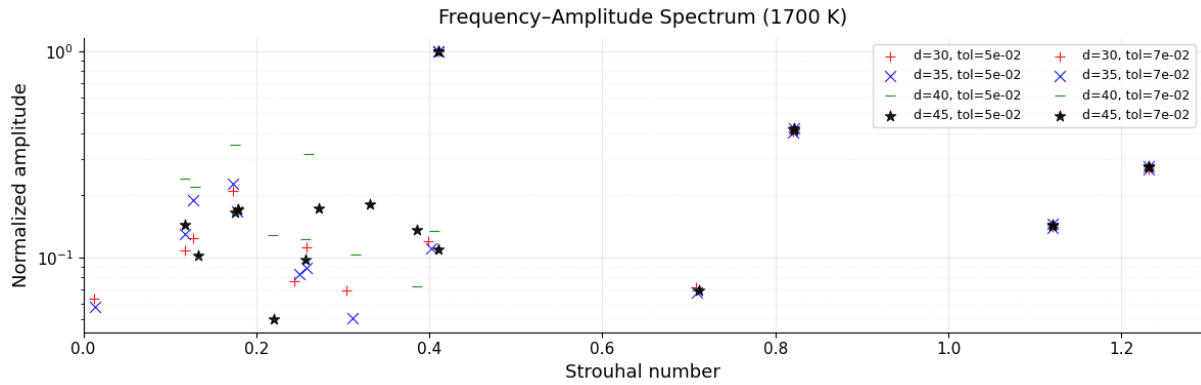
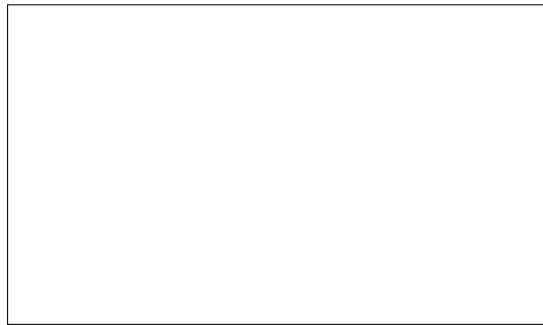
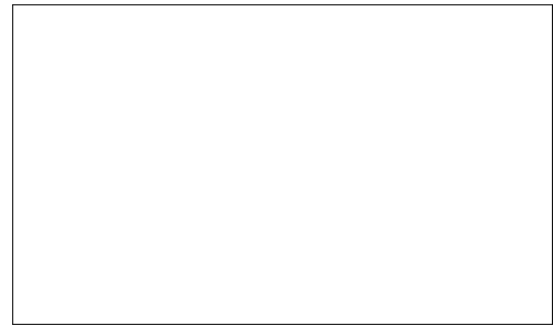


Figure 6: Normalised DMD mode amplitudes as a function of Strouhal number for the 1700 K dataset, for $d \in \{35, 40, 45\}$ and $\varepsilon_{\text{tol}} \in \{5 \times 10^{-2}, 7 \times 10^{-2}\}$.

The associated modes to the highlighted points in Figure 6 are shown in Figure 7.



(a) [PLACEHOLDER] Mode 1



(b) [PLACEHOLDER] Mode 2



(c) [PLACEHOLDER] Mode 3



(d) [PLACEHOLDER] Mode 4

Figure 7: [PLACEHOLDER] DMD modes associated with the dominant peaks in the 1700 K spectrum.

Discussion: Spectral and Spatial Evolution Across Preheat Temperatures

The three preheat temperature cases investigated here represent distinct combustion regimes with direct relevance to the efficiency–pollutant trade-off inherent in NH_3/H_2

combustion. The 1200K case, in practical application has lower thermal efficiency, while the high temperature one, i.e. 1700 K is overall more efficient due to a more complete combustion but at the cost of significantly enhanced NO_x formation. The 1500 K case sits at the boundary between these two regimes and is therefore of particular practical interest as a candidate operating point for studying the tradeoff between combustion efficiency and pollutant emissions. In particular we focused on understanding the behaviour of the flow rather than the chemical one.

At 1200 K the structure of the flow found by the algorithm is harmonic, with a dominant peak at $St \approx 0.42$ is accompanied by secondary peaks at approximately integer multiples of this frequency extending up to $St \approx 1.7$. This is the sign of a strongly periodic shear layer instability, in which the primary roll up vortex and its successive pairings organize the flow in a regular pattern.

On the other hand, at 1700 K, the spectral picture changes substantially. While the dominant frequency is preserved at $St \approx 0.40$, confirming that the primary instability mechanism remains kinematically controlled by the shear between the two inlet streams, the harmonic content is severely reduced, and the overall spectrum is sparser. This evolution is physically consistent with the effect of increased thermal loading: the elevated heat release rate and associated gas expansion at 1700 K tend to suppress vortex pairing and subharmonic modes, leading to a quasi-periodic, more turbulent flow.

The 1500 K case contains the transition between the two cases. In this spectrum we can find dominant harmonic structures as in the 1200 K, but additionally the creation of quasi-periodic frequencies that are not in a neighborhood of the main one. This quasi-periodic spectral content can be interpreted as a signature of the competing dynamics active at this operating point: the primary shear-layer instability is still present, but thermal effects are beginning to modify the nonlinear interactions.

The coherent structures corresponding to the main physical frequencies identified provide further information about the dynamics of the phenomenon. At 1200 K, the modes with highest amplitude are spatially extended structures concentrated in the far field of the domain as shown in the Figure DAMETTERE, reflecting the downstream convection and roll-up of large-scale vortices that have had space and time to develop along the mixing layer. At 1700 K the dominant spatial activity is shifted towards the near field, immediately downstream of the splitter plate. This can be explained by the fact that the stronger heat release ties the energetically significant dynamics to the flame attachment as shown in Figure DAMETTERE with less coherent large-scale activity further downstream. The transient case, i.e. 1500 K has energetically significant DMD modes in both the far field and the near field regions as shown in the DMD modes associated to the frequencies DAMETTERE and the DMD MODES DAMETTERE. In this regime the instabilities are affected by both the far field convective instability and the energy release near the inlet.

5 Conclusions

The algorithm proposed has identified that the preheat temperature can be seen as a modulatory parameter on the shear-layer instability. The increase in temperature suppresses the secondary harmonics and shift the dynamics of the instability from the far field to the near one. The speactral profile of the flow changes from periodic to quasi periodic. Further investigation of the 1500 K operating point, through parametric studies and refined modal decomposition techniques, may enable the identification of conditions that simultaneously minimise NO_x emissions and maximise combustion efficiency



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